Credit Risk Portfolio Analytics – Solution Overview

Basel Scorecard, Explainable AI Systems & Production Onboarding

Scope: Retail & Commercial Loan Portfolio (14,685 accounts)
Default Rate: 16.10% | **Target:** 90+ DPD Delinquency

1. DASHBOARD CAPABILITIES & RISK ANALYST WORKFLOW

View 1: Portfolio Macro View

Purpose: Executive oversight of portfolio health and capital at risk ($228.4M exposure).

- Dynamic filters across product types, geographic regions, and employment sectors.
- Displays risk tier migration (Low: <10%, Med: 10-30%, High: >30% PD).

View 2: Account Deep Dive

Purpose: Single-borrower loan underwriting & forbearance review.

- Interactive risk speedometer displaying real-time calibrated default probability.
- Top 3 local risk drivers (e.g., Shortfall +$1,200, Bureau DPD 90) derived via standardized coefficients.

View 3: Ask the Portfolio (AI)

Purpose: Natural language executive ad-hoc portfolio intelligence.

- Allows analysts to ask complex cross-table questions in plain English without SQL.
- Backed by zero-hallucination deterministic Python math with source column lineage.

View 4: Anomaly Review Queue (AI)

Purpose: Operational early-warning surveillance queue.

- Surfaces accounts exhibiting high-risk behavioral divergences before 90-day write-off.
- Enables human-in-the-loop audit actions: *Confirm Anomaly* or *Dismiss Alert*.

2. OPERATIONAL AI GOVERNANCE SPECIFICATIONS

AI Feature	Operational Trigger	False Positive Scenario	False Negative Scenario	Human Review Step
AI 1: Anomaly Detector Rule-Based Vectorized	1. Sudden DPD jump >30 with clean past. 2. Shortfall >2.5 SD of loan cohort. 3. Clean on-us but severe bureau default.	Borrower experiences payroll delay or holiday clearing friction, curing within 5 days.	Strategic defaulter paying micro-installments just high enough to stay under 2.5 SD limit.	Credit officer inspects peer deviation, verifies collateral, and routes to collections or dismisses.
AI 2: Ask Portfolio Grounded LLM Engine	Risk analyst submits natural language portfolio query in UI.	Query matches intent category (e.g., region) when user requested cross-tabulation.	Freeform phrasing not caught by deterministic intents (system triggers safe fallback).	Source columns and numerical proofs are displayed alongside output for instant verification.

3. SCORECARD METHODOLOGY & RESULTS

0.6727	26.13%	0.3455
ROC-AUC	KS STATISTIC	GINI INDEX

Regulatory Champion: Balanced Logistic Regression was selected over tree ensembles (Gradient Boosting AUC: 0.6509). Logistic regression achieved superior rank ordering, zero black-box opacity, and straightforward coefficient explainability required under Basel II/III model governance.

4. TOP 3 PORTFOLIO STRATEGIC INSIGHTS

- **Unsecured Product Stress:** Personal Loans and Credit Cards exhibit default rates exceeding **21.5%**—more than 2.8x higher than Collateralized Mortgages (7.8%).
- **Employment Vulnerability:** Unemployed borrowers carry a **24.2% default rate**. Debt-to-income caps must be enforced for gig/unemployed segments.
- **External Bureau Spillover:** Prior external write-offs increase default probability by **78%**, proving on-us data alone misses hidden distress.

5. KEY ASSUMPTIONS & BANK IT / DATA PRODUCTION REQUIREMENTS

Core Modeling Assumptions:

- Missing bureau DPD signifies an active account in good standing (0 DPD).
- Historical macroeconomic stability; stationarity across loan cohorts.
- Repayment shortfall is calculated against scheduled EMI obligations.

Requirements from Bank IT & Data Engineering:

- **Automated Data Pipelines:** Daily batch ETL ingestion via Airflow/Databricks to refresh child aggregations and eliminate Cartesian joins.
- **Feature Store & Model Registry:** MLflow model registry for versioning model weights and tracking quarterly population drift (PSI).
- **Enterprise LLM Gateway:** Secure internal endpoint (Azure OpenAI / private API gateway) with PII masking before sending data to LLM.